



Streamlining Automated Laboratory Workflows in Tightly Firewalled Networks

Eli Fine, PhD

Society for Laboratory Automation & Screening - Boston, MA

February 10, 2026

Firewalls are Closing In

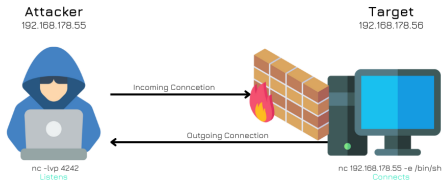
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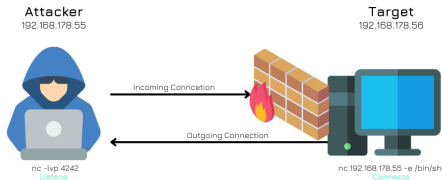
How a Reverse Shell works



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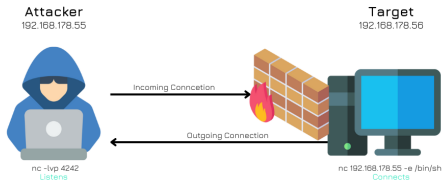
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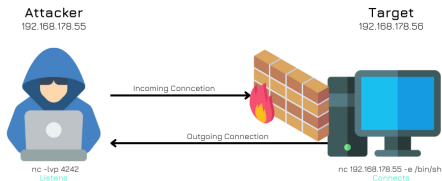
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- Automation inroads into Manufacturing; tighter restrictions or true air-gap
- Even company-hosted services sometimes blocked by 'catch-all' IT policies

How a Reverse Shell works



Challenges with Notifications/Alerting

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- Many disparate lab systems/software to update to match new approach



Challenges Using a Simulation Environment

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- Vendor tools to 'snapshot' workcells become cumbersome at scale
- Access to cloud-based version control (e.g. GitHub) is blocked

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- Cloud-based SaaS or corporate Git servers may be blocked by the firewall
- Many automation users unfamiliar with Git

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 - Be user-friendly for people with limited experience using Git
- Be compatible with Windows 10/11 (and Linux)

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- Open-source, to facilitate security scanning & review
- Prefer containerized applications when possible to enhance compatibility
- Avoid Docker Desktop licensing concerns
- Browser-based UIs to be accessible from anywhere on the network

Setting up Host Computer to run Containers

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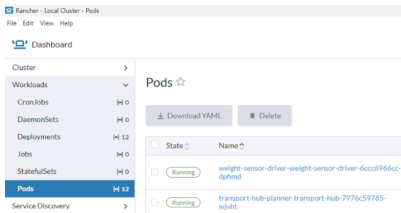
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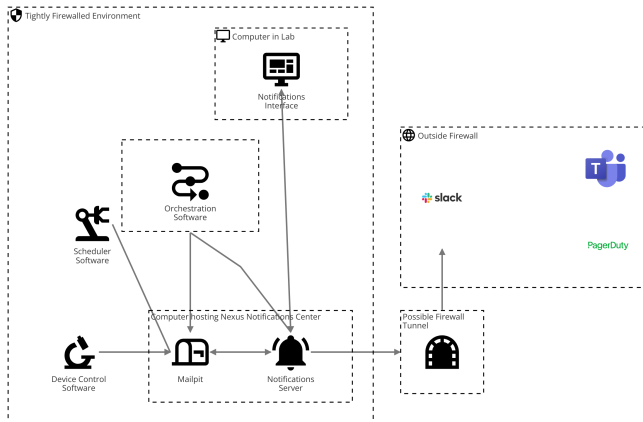
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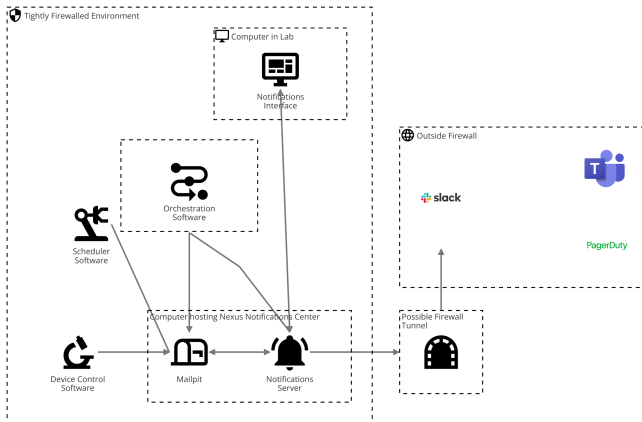


Observability Architecture



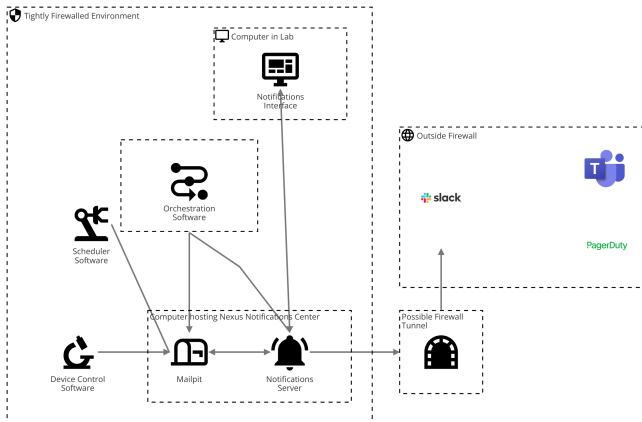
Observability Architecture

- Mailpit provides an email address to receive notifications



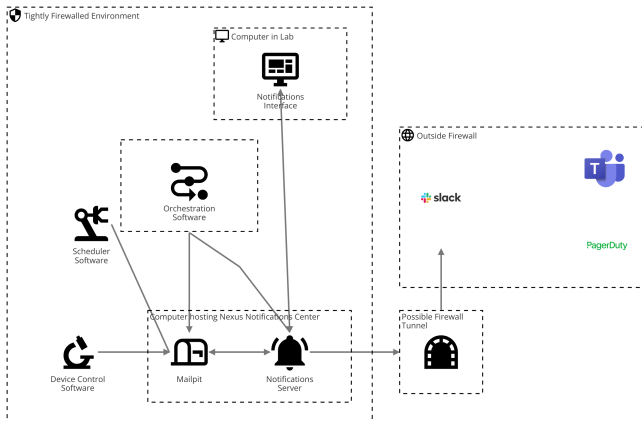
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- Notifications can be sent via email or directly via API calls to the Nexus Notifications Center
- User interface accessible anywhere in lab



Observability Dashboard — Nexus Notifications Center

LAB SYNC

Dashboards

📧
Emails

🔔
Custom Alerts

⚙️
Settings

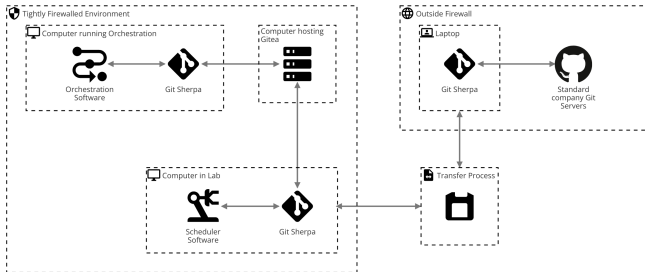
Nexus - Notifications Center

Emails

🔼🔽 Subject	📧 Sender	👤 Recipient	🕒 Timestamp
Warning: High failure rate for protocol 'cDNA_synthesis'	analytics@lab-lab.com	science-leads@lab-lab.com	Thu Feb 05 2026 06:23:52 GMT-0500 (Eastern Standard Time)
Error: Pipetting robot R-12 failed tip pickup	automation@lab-lab.com	ops@lab-lab.com	Thu Feb 05 2026 06:23:52 GMT-0500 (Eastern Standard Time)
Tip Waste Full	flow-cytometry-workcell@analytics-lab.local	glbg-alerts@analytics-lab.local	Wed Feb 04 2026 17:50:26 GMT-0500 (Eastern Standard Time)
Warning: Freezer-2 temperature high	freezer-monitor@lab-lab.com	facilities@lab-lab.com	Thu Feb 05 2026 06:23:52 GMT-0500 (Eastern Standard Time)
Info: Reagent delivery received	logistics@lab-lab.com	inventory@lab-lab.com	Thu Feb 05 2026 06:23:52 GMT-0500 (Eastern Standard Time)
Info: Maintenance window scheduled	maintenance@lab-lab.com	infra@lab-lab.com	Thu Feb 05 2026 06:23:52 GMT-0500 (Eastern Standard Time)
Error: Robot collision detected	orchestrator@lab-lab.com	ops@lab-lab.com	Thu Feb 05 2026 06:23:52 GMT-0500 (Eastern Standard Time)
Info: Plate scan complete	qc@lab-lab.com	team-genomics@lab-lab.com	Thu Feb 05 2026 06:23:52 GMT-0500 (Eastern Standard Time)
Error: Tip rack misaligned	robot@lab-lab.com	ops@lab-lab.com	Thu Feb 05 2026 06:23:52 GMT-0500 (Eastern Standard Time)
Info: Scheduled run started	scheduler@lab-lab.com	team-genomics@lab-lab.com	Thu Feb 05 2026 06:23:52 GMT-0500 (Eastern Standard Time)
Warning: Reagent low - MasterMix MM-2026-01	scheduler@lab-lab.com	inventory@lab-lab.com	Thu Feb 05 2026 06:23:52 GMT-0500 (Eastern Standard Time)

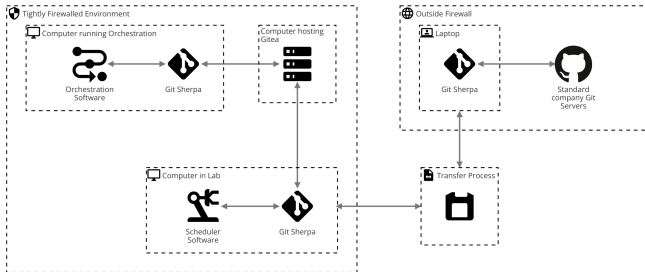
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Version Control Architecture



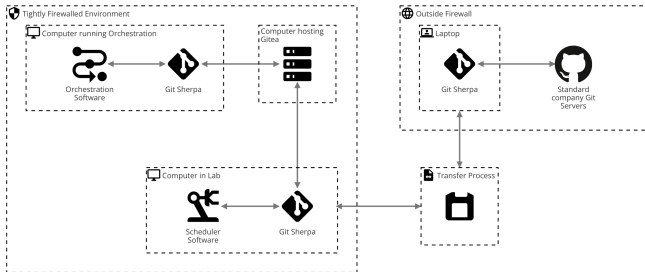
Version Control Architecture

- Gitea runs as a Git Server within the lab



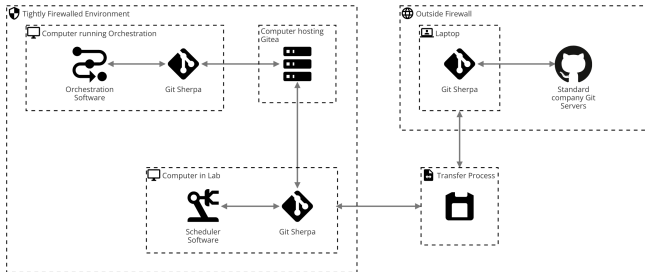
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- 'Git Sherpa': purpose-built app to guide users through Git

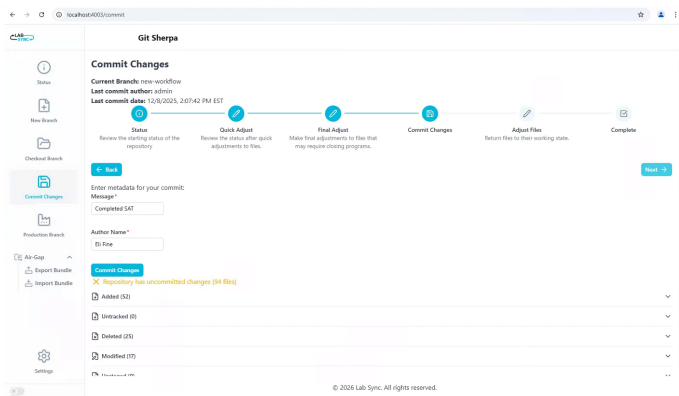


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- Git Bundles move changes across the boundary



Git Sherpa — Guiding Users through Git



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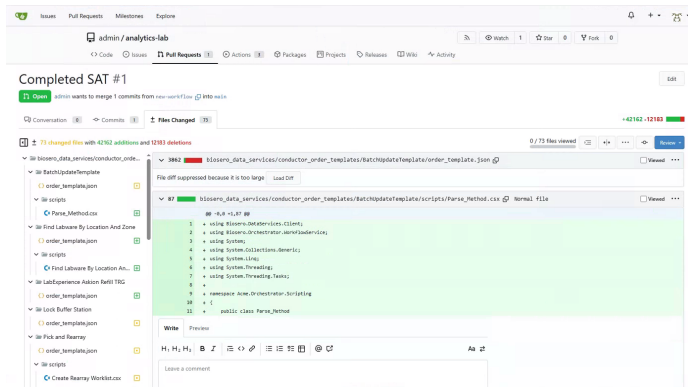
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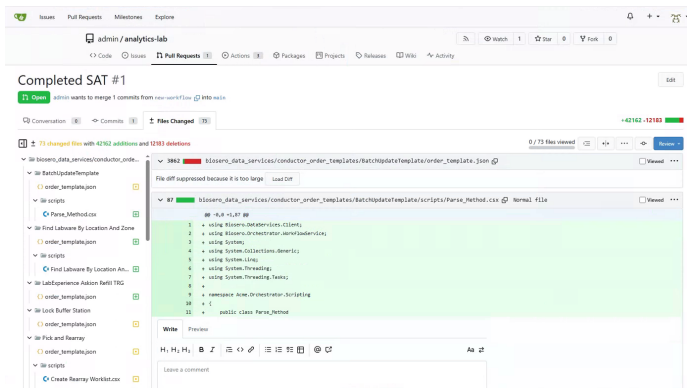
```
- relative_path_in_repo: gbg_instances/rearray/documents/Settings/Settings.xml
  xml_adjustments:
  - xpath_query: //ns:KeyValueOfstringanyType[ns:Key='DataServices.IP']/ns:Value
    desired_value_sources:
      - environmental_variable_name: BIOSERO_GBG_DATA_SERVICES_URL
      - string_value: http://localhost:8105/api/v2.0/
```

Gitea — Fully Featured Git in the Lab



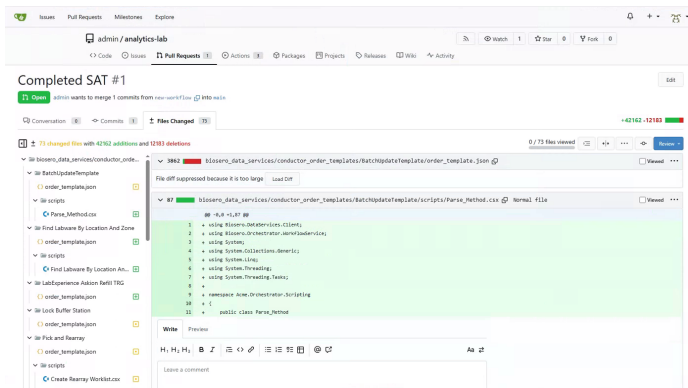
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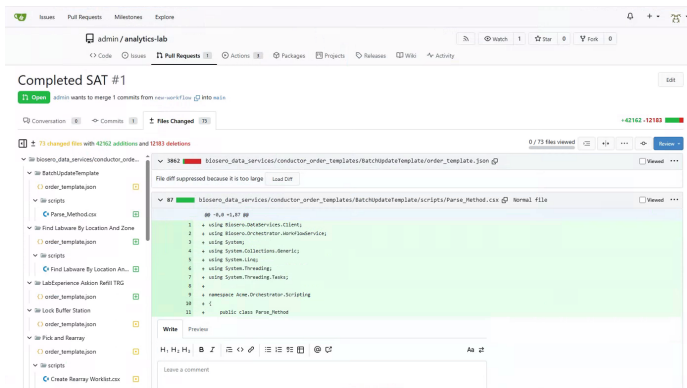
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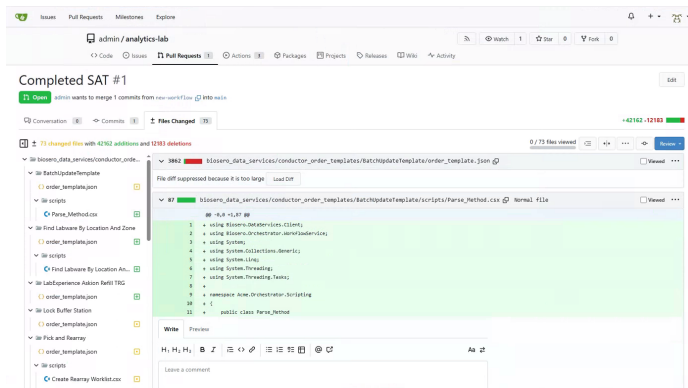
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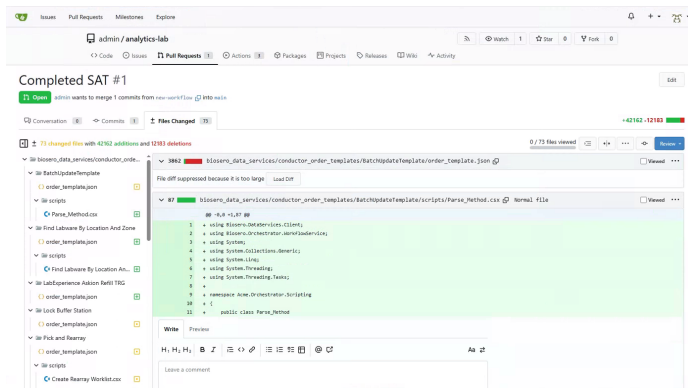
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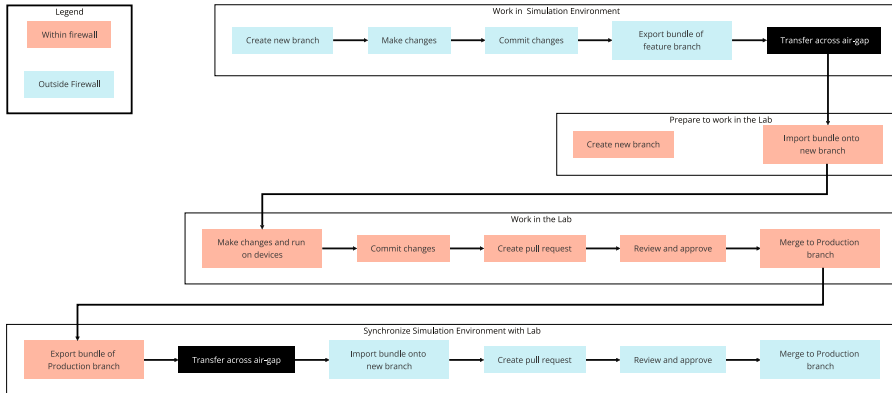
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- Containerized applications can recreate familiar functionality within firewall
- Nexus Notifications Center can aggregate alerts via email and API calls
- Git Sherpa makes version control approachable with minimal friction
- Combining Git Sherpa, Gitea, and Bundles enables version control in the lab synchronized with a simulation environment
- Open source approach allows community contributions to expand Git Sherpa's ability to version control additional lab automation software necessitating 'custom' interactions

Questions?

Version Control Across the Air-gap — Workflow



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- Create the ability for users to create their own "plugins" for Git Sherpa for placing unusual pieces of lab automation configuration under version control
- Enable users to create "plugins" for how Nexus Notifications Center forwards alerts to a system within the firewall or through a tunnel to external services